Network - Mid Credit ~~2025~~ 2026

1. The main difference that distinguishes HTTP/2 over the other earlier versions is ......
(A) The server responds in order (FCFS) to GET requests. → HTTP1
(B) The connection is persistent.
HTTP3 ← (C) It adds security, per-object error, and congestion control (more pipelining) over UDP.
(D) Transmission order of requested object based on client-specified object priority. → HTTP2

2. ...... is a formal document outlining technical specifications, organizational notes, and standards relevant to internet and networking technologies, including routing, addressing, and transport technologies.
(A) ISP
(B) RFC
(C) DELAM
(D) IETF

3. What is the purpose of DNS in a computer network?
(A) To convert domain names to IP addresses
(B) To encrypt data transmission
(C) To manage network security
(D) To establish wireless connections

4. The Protocol field in the IPv4 header indicates ......
(A) The transport layer protocol used in the data portion
(B) The type of network layer protocol
(C) The type of data link layer protocol
(D) The routing algorithm used

IPv4 header
↳ transport layer

5. ...... provides retrieval and deletion of the stored E-mail messages on the server.
(A) IMAP
(B) SMTP
(C) POP
(D) HTTP

6. In Ethernet, the 'Type' field value 0x0800 indicates that the frame carries ......
(A) ARP packet
(B) IPv6 packet
IPv4 (C) IPv4 packet
(D) RARP packet

800 ↳ 4   0x800 ↳ 4

7. The typical transmission rate for wide-area cellular access networks is ......
(A) 1 Gbps
(B) 100 Kbps
(C) 10's Mbps
(D) 10's Gbps

8. What is the minimum size of an Ethernet frame (excluding preamble and SFD)?
(A) 46 bytes
(B) 1500 bytes
(C) 64 bytes
(D) 72 bytes

64 → 64

9. What is the advantage of a tree network topology?
(A) Point-to-point wiring for individual segments
(B) Direct connectivity between devices without relying on a central device
(C) Easy to install and wire
(D) Scalability for future network expansion

10. ...... is the time for a small packet to travel from client to server and back.
(A) Internet delay
(B) RTT
(C) Access link delay
(D) LAN delay

11. Which mode of operation involves the output of data from the sender being received by all other receivers?
(A) Simplex
(B) Broadcast
(C) Half Duplex
(D) Full Duplex

12. ...... cable is two concentric copper conductors that can work in bidirectional with multiple frequency channels on cable.
(A) Coaxial
(B) Fiber optic
(C) Twisted pair category 5
(D) Twisted pair category 6

13. Which organization is responsible for international LAN standards?
(A) IETF
(B) IEEE
(C) ISO
(D) ITU

S P F
A P S T N D P
seg. TCP/UDP MAC error

14. What are the services provided by the data link layer?
(A) Framing, error detection and correction, flow control
(B) Logical addressing, routing
(C) Segmentation reassembly, error checking
(D) Session establishment, maintenance, synchronization

15. How many bits are used to represent the network portion of an IP address in Class A?
(A) 24 bits
(B) 32 bits
(C) 8 bits
(D) 16 bits

Class A

16. Which standard is commonly used for Gigabit Ethernet over copper wire?
(A) 1000BaseSX
(B) 1000BaseLX
(C) 1000BaseCX
(D) 1000BaseT

EtherneT

17. Communication is the main component of the network core that concerns the network interconnection.
(A) Server
(B) Host
(C) Hub
(D) Router

18. What is the standard number for Token Ring technology defined by the IEEE?
(A) 802.3
(B) 802.11
(C) 802.5

(D) 802.8

19. Which IP addressing method allows for variable-length subnet masks?
(A) Classful addressing
(B) Classless addressing
(C) Class-based addressing
(D) Supernetting

20. What type of model does HTTP follow?
(A) Client-server model
(B) Peer-to-peer model
(C) Broadcast model
(D) Hybrid model

21. Which of the following fields comes first in an Ethernet frame?
(A) Destination MAC Address
(B) Type / Length
(C) Source MAC Address
(D) Frame Check Sequence (FCS) → last

22. What is the size of the IPv4 header without any options?
20
(A) 46 bytes
(B) 64 bytes
(C) 20 bytes
(D) 32 bytes

23. UDP is suitable for applications that prioritize:
(A) Low latency and real-time data transmission
(B) Reliability
(C) Flow control
(D) Ordered data transmission

24. What is the purpose of a loopback address in IP addressing?
(A) To identify the network to which a device belongs
(B) To encrypt data transmitted over the network
(C) To assign unique IP addresses to devices
(D) To test network connectivity on a device

25. Communication provides encrypted TCP connections, data integrity, and end-point authentication.
(A) TCP sockets
(B) TLS
(C) Vanilla TCP & UDP sockets
(D) UDP sockets

TCP UDP

26. What are the two types of packet-switched networks?
(A) Broadcast and Multicast
(B) Simplex and Duplex
(C) Circuit Switching and Packet Switching
(D) Connection-oriented and Connectionless

27. What does the prefix of an IP address identify?
(A) The host within the network
(B) The network to which the computer is connected
(C) The geographical location of the computer
(D) The protocol used for communication

Final 2025 (General)

Q1. _______ provides retrieval and deletion of stored e-mail messages on the server.
(A) IMAP (B) SMTP (C) POP (D) HTTP

Q2. _______ method sends user input from client to server in the entity body of an HTTP request.
(A) GET (B) POST (C) HEAD (D) PUT

Q3. In _______ control plane, a remote controller computes and installs forwarding tables in routers.
(A) per-router (B) DSL (C) QoS (D) SDN

Q4. In _______, the receiver sends ACKs for correct packets and NAKs for corrupted packets.
(A) rdt2.1 (B) rdt2.2 (C) rdt2.0 (D) rdt3.0

rtd 2.0 < correct / corrupted
" 2.1 < 0 / 1 seq. number
2.2 → free NAK
3 → error & loss

Q5. All of the following are DNS services, except _______.
(A) Centralized hostname-to-IP translation
(B) Host aliasing
(C) Mail server aliasing
(D) Load distribution

Q6. In _______, voice and data are transmitted at different frequencies using _______.
(A) DSL, DSLAM
(B) Cable-based access, DSLAM
(C) Cable-based access, CMTS
(D) DSL, CMTS

Q7. Using _______, a host dynamically obtains an IP address when joining a network.
(A) DHCP (B) CIDR (C) WFQ (D) NAT

Q8. _______ provide(s) no encryption and transmit passwords in cleartext.
(A) TLS
(B) TCP sockets
(C) UDP sockets
(D) Vanilla TCP & UDP sockets

Q9. In _______, only ACKs are used instead of NAKs.
ACK / NAK > (A) rdt2.0 (B) rdt2.2 (C) rdt3.0 (D) rdt2.1 < 0 / 1

Q10. All are reflections of best-effort service, except _______.
(A) Simplicity enables wide deployment
(B) Sufficient bandwidth supports real-time apps
(C) Distributed services near clients
(D) Guaranteed delivery with < 40 ms delay

Q11. The UDP segment header size is _______ bytes.
(A) 24 (B) 16 (C) 32 (D) 20

Q12. A _______ process waits to be contacted.
(A) Client (B) P2P (C) Server (D) Inter

Q13. _______ is a group of device interfaces that can reach each other without a router.
(A) Subnet (B) WFQ (C) NAT (D) DHCP

Q14. All are features of UDP, except _______.
(A) No connection establishment
(B) No connection state
(C) Large header size → small
(D) No congestion control

Q15. _______ supports flow control and window size management.
(A) Flow controlled
(B) Connection oriented
(C) Pipelining
(D) Full duplex dialog

Q16. In _______, end-to-end resources are reserved between source and destination.
(A) Packet switching
(B) Routing
(C) Forwarding
(D) Circuit switching

Q17. Each HTTP request header line ends with _______.
(A) Carriage return and line feed
(B) Cabin and line feed
(C) Carriage return and semicolon
(D) Carriage return

Q18. In a home network, all are often combined in one box, except _______.
(A) Cable or DSL modem
(B) Router, firewall, NAT
(C) WiFi access point
(D) Wireless and wired devices

Q19. All are network edge devices, except _______.
core
(A) Server
(B) Mobile phone
(C) Car
(D) Router

Q20. _______ uploads and replaces a file at a specified URL.
(A) POST (B) GET (C) HEAD (D) PUT

Q21. All are network defense mechanisms, except _______.
(A) Access restrictions
(B) Authentication
(C) DNS
(D) Firewalls

Q22. What is the queuing delay for the packet?
(A) 19.2 ms (B) 21.6 ms (C) 16.8 ms (D) 2.4 ms

Q23. The key feature of HTTP/2 is _______.
(A) FCFS server response → HTTP 1
(B) Persistent connection
(C) UDP-based transmission
(D) Client-specified object priority → HTTP 2

Q24. Websites use _______ to maintain state between transactions.
(A) Cookies
(B) Sessions
(C) Databases
(D) Web caches

Q25. _______ cable carries light pulses and is immune to electromagnetic noise.
(A) Fiber optic

(B) Coaxial
(C) Twisted pair Cat 6
(D) Twisted pair Cat 5

Q26. _______ define message formats, order, and actions in communication.
(A) Service
(B) ISP
(C) Protocol
(D) Router

Q27. Typical transmission rates for Cat 5 cable are _______.
(A) 10 Mbps, 100 Mbps
(B) 100 Mbps, 1 Gbps
(C) 10 Gbps, 100 Gbps
(D) 10 Gbps

Q28. All are services for a datagram flow, except _______.
(A) Guaranteed delivery
(B) In-order delivery
(C) Guaranteed minimum bandwidth
(D) Inter-packet spacing control

Q29. In DNS NS records, the name is _______ and the value is _______.
(A) Hostname, IP address
(B) Alias name, canonical name
(C) Hostname, SMTP server name
(D) Domain, authoritative name server hostname

Q30. _______ layer handles encryption, compression, and data representation.
(A) Application
(B) Presentation
(C) Session
(D) Data link

Q31. All are features of SMTP, except _______.
(A) ASCII command/response
(B) Client pull
(C) Multipart messages
(D) CRLF.CRLF message termination

Mid General 26

1. The Frame Check Sequence (FCS) field in an Ethernet frame is used for

A. Error correction
B. Flow control
C. Error detection using CRC
D. Frame synchronization

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2. The non-persistent HTTP response time equals to

A. 2RTT
B. as little as one RTT for all the referenced objects
C. as little as two RTT for all the referenced objects
D. 2RTT + file transmission time

---

3. What is the default port for FTP?

A. 21
B. 25
C. 53
D. 80

---

4. What is the purpose of the Header Checksum field in the IPv4 packet?

A. To specify encryption type
B. To detect errors in the header
C. To identify packet fragments
D. To verify the integrity of the data field

---

5. HTTP uses port number ___80___ whereas SMTP uses ______

A. 80, 21
B. 80, 20
C. 25, 80
D. 80, 25

---

6. ______ is commonly used in traditional telephone networks.

A. circuit switching
B. routing
C. packet switching
D. forwarding

---

7. Which field in an IPv4 header specifies the length of the entire packet, including header and data?

A. Fragment Offset
B. Header Length
C. Total Length
D. Identification

---

8. Which IPv4 header field is used to prevent packets from circulating indefinitely in the network?

A. Protocol
B. Time to Live (TTL)
C. Fragment Offset
D. Header Checksum

---

9. In ______ a promiscuous network interface reads/records all packets passing by.

A. packet sniffing
B. Malware
C. IP spoofing
D. Dos

---

10. Which protocol supports packets containing error messages, control, and information?

A. ICMP
B. ARP
C. UDP
D. RARP

---

11. Which technology uses a fixed packet size of 53 bytes?

A. Satellite systems
B. DQDB
C. Ethernet
D. ATM

---

12. Which IP address class is used for multicast?

A. Class B
B. Class C
C. Class A
D. Class D

multi → الأخير → D

---

13. What does a distance vector algorithm assign to each link in the network?

A. Cost numbers
B. Bandwidth values
C. IP addresses
D. MAC addresses

---

14. What are the four phases of DHCP communication known as?

A. IPAD
B. DORA
C. DHCP
D. ABCD

DHCP phases → DORA

الـDHCP هاجبة DORA

---

15. What is the purpose of NAT (Network Address Translation)?

A. To translate domain names to IP addresses
B. To encrypt network traffic
C. To map private IP addresses to public IP addresses
D. To assign IP addresses to devices

---

16. Which protocol is used by the Novell Netware operating system for data transfer within networks?

A. IPX/SPX
B. RARP
C. ICMP
D. ARP

---

17. What is the maximum payload size allowed in a standard Ethernet frame?

A. 2048 bytes
B. 1024 bytes
C. 512 bytes
D. 1500 bytes

---

18. Which wireless standard provides the highest data transfer rate?

A. 802.11b
B. 802.11a
C. 802.11g
D. 802.11ac

---

19. What does the throughput rate represent in a network?

A. The capacity of the network's servers and workstations
B. The speed at which information can be sent between devices
C. The highest and lowest frequencies in the network
D. The rate of incoming data at a specific point in the network

---

20. The typical transmission rates for wireless local area networks are

A. 11, 56, 540 Mbps
B. 11, 54, 450 Kbps
C. 11, 54, 450 Mbps
D. 11, 56, 540 Kbps

---

security

21. The main difference that distinguishes HTTP/3 over earlier versions is

A. The connection is persistent.
B. Transmission order of requested objects based on client-specified object priority.
C. It adds security, per object error- and congestion control (more pipelining) over UDP.
D. The server responds in order (FCFS) to GET requests.

---

22. Which protocol is used for retrieving emails from a mail server?

A. SMTP
B. POP3
C. FTP
D. IMAP

---

23. What happens when data packets collide in CSMA/CD?

A. The packets are discarded
B. The packets wait for a random time before retransmission
C. The packets are combined into a single packet
D. The packets are retransmitted immediately

---

24. ______ permit sites to learn a lot about the user on their site.

A. Sessions
B. Web caches
C. Databases
D. Cookies

---

25. What does MAN stand for?

A. Metropolitan Area Network
B. Metropolitan Access Network
C. Mobile Area Network
D. Multiple Area Network

---

26. Which layer handles the formatting, encryption, and compression of data in the OSI model?

A. Data link layer
B. Presentation layer
C. Session layer
D. Physical layer

---

27. How many bits are used to represent the host portion of an IP address in Class B?

A. 24 bits
B. 8 bits
C. 32 bits
D. 16 bits

---

28. ______ is a standards organization for the Internet and is responsible for the technical standards that make up the Internet protocol suite (TCP/IP).

A. ISP
B. CMTS
C. IETF
D. RFC

---

29. ______ method includes user data in the URL field of the HTTP request message following a '?'.

A. PUT
B. GET → تجيب الداتا
C. HEAD
D. POST

---

30. ______ different channels transmitted in different frequency bands.

A. TDM
B. FDD
C. TDD
D. FDM

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Quiz 01 (5 Marks) - Model (1)

Student Name: .... ID:................................

Choose the right answers and label them in your answer sheet (Q1-Q10 x 0.5 Mark)

1. .... provide(s) no encryption and cleartext passwords sent into socket traverse Internet in cleartext.
   (A) TLS (B) TCP sockets (C) UDP sockets **(D) Vanilla TCP & UDP sockets**

clear no enc.] Vani.

2. .... manages root DNS domain. root → ICANN
   (A) ICANN (B) DNSSEC (C) Comcast (D) Akamai

3. The three major components of the E-mail are the following, except ....
   (A) user agents **(B) IMAP** (C) mail servers (D) SMTP

4. The persistent HTTP response time equals to ....
   (A) 2RTT+ file transmission time **(B) as little as one RTT for all the referenced objects** (C) as little as two RTT for all the referenced objects (D) 2RTT

5. In DNS records, if the type is MX, the name is ...., and the value is .... Mr.X → host
   (A) domain, the hostname of the authoritative name server **(B) hostname, name of SMTP mail server associated with it** (C) hostname, IP address (D) alias name, canonical name

6. UDP can be used for all of the following, except ....
   (A) DNS (B) HTTP/3 (C) SNMP **(D) IMAP**

7. In ...., demultiplexing uses 4-tuple: source and destination IP addresses and port numbers.
   (A) UDP (B) IP **(C) TCP** (D) Socket → reliable

8. Users are generating data at a rate of 100 kbps when busy, but are busy generating data only with probability p = 0.1. Suppose that we have 1 Gbps link What is N, the maximum number of users that can be supported simultaneously under circuit switching?
   (A) 10000 (B) 1000 (C) 5000 (D) 100000

10000

9. .... uses header info to deliver received segments to the correct socket.
   (A) Multiplexing **(B) Demultiplexing** (C) Routing (D) Forwarding

10. In ...., it handles garbled ACK/NAKs by adding a sequence number to the pkt.
   (A) rdt2.0 **(B) rdt2.1** (C) rdt2.2 (D) rdt3.0

0, 1

No.	Answer	No.	Answer	No.	Answer	No.	Answer	No.	Answer
1		7		13		19		25	
2		8		14		20		26	
3		9		15		21		27	
4		10		16		22		28	
5		11		17		23		29	
6		12		18		24		30	

The end of the examination
Best Wishes
Prof. Dr. Mohammed Elmogy

$\text{rate} = 100\text{kbps} = 100 \times 10^3 \text{ bps} = 10^5$

$1\text{Gbps} = 10^9$

$\frac{10^9}{10^5} = 10^4 = 10000$

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Quiz 01 (5 Marks) - Model (2)

Choose the right answers and label them in your answer sheet (Q1-Q10 x 0.5 Mark)

1. Once any name server learns mapping, it caches mapping and immediately returns a cached mapping in response to a query. It will do the following, except ....
   (A) Caching improves response time (B) Cache entries timeout after some time (TTL) (C) TLD servers are typically cached in local name servers **(D) Average response time equals 2RTT**

2. Each header line in the HTTP request message should end with ....
   (A) carriage return and line feed (B) colon and line feed (C) carriage return and semi-colon (D) carriage return

3. .... process waits to be contacted.
   (A) Client (B) P2P **(C) Server** (D) Inter

4. .... method is used when the web page often includes form input and the user input sent from client to server in the entity-body of HTTP request message.
   (A) GET **(B) POST** (C) HEAD (D) PUT

   POST تبعت

5. All of the following are the main features of SMTP, except ....
   (A) Each object encapsulated in its response message (B) It uses persistent connections (C) Client push (D) It requires the message (header and body) to be in 7-bit ASCII

6. Consider sending over HTTP/2 a Web page that consists of one video clip, and five images. Suppose that the video clip is transported as 2000 frames, and each image has three frames. If all the video frames are sent first without interleaving, how many "frame times" are needed until all five images are sent?
   (A) 2000 (B) 2030 (C) 4030 **(D) 2015**

7. In ...., the underlying channel can have errors and lose packets (data, ACKs).
   (A) rdt2.2 (B) rdt2.0 **(C) rdt3.0** (D) rdt2.1 0 1 seq

8. The UDP segment header contains the following information .... ACK NAK
   (A) Source IP address, Source port number, Destination IP address, Destination port number (B) Source port number, Destination port number, Seq. Number, Length **(C) Source port number, Destination port number, Length, Checksum** (D) Source IP address, Source port number, Destination IP address, Destination port number, Length, Checksum

9. A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2.5 Mbps. What is the queuing delay for the packet?
   $R = 2.5 \times 10^6$ 4.5 $L = 1500 \times 8$
   (A) 19.2 ms **(B) 21.6 ms** (C) 16.8ms (D) 2.4 ms $\frac{L}{R} \times 4.5 = 0.216\,S$
   $= 21.6ms$

10. In ...., demultiplexing using destination port number only.
    (A) UDP (B) TCP (C) IP (D) Socket

No.	Answer	No.	Answer	No.	Answer	No.	Answer	No.	Answer
1		2		3		4		5	
6		7		8		9		10	

The end of the examination
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Quiz 03 (5 Marks) - Model (3)

Choose the right answers and label them in your answer sheet (Q1-Q10 x 0.5 Mark)

1. All of the following are the main features of SMTP, except ....
   (A) It has an ASCII command/response interaction and status codes **(B) Client pull** (C) Multiple objects sent in a multipart message (D) The server uses CRLF.CRLF to determine the end of the message

2. .... process initiates communication.
   (A) P2P (B) Server (C) Inter **(D) Client**

   Domain

3. In DNS records, if the type is NS, the name is ...., and the value is ....
   (A) hostname, IP address (B) alias name, canonical name (C) hostname, name of SMTP mail server associated with it **(D) domain, the hostname of the authoritative name server**

4. .... satisfy client requests without involving the origin server.
   (A) Cookies (B) Sessions **(C) Web caches** (D) Databases

5. .... method uploads a new file or object to the server. It also completely replaces files at specified URLs with the content in the entity-body of the POST HTTP request message.
   (A) POST (B) GET (C) HEAD **(D) PUT**

6. .... handles data from multiple sockets and adds a transport header.
   (A) Multiplexing (B) Demultiplexing (C) Routing (D) Forwarding

7. In TCP, the sender will not overwhelm the receiver. It is called ....
   (A) connection-oriented (B) pipelining (C) full duplex data **(D) flow controlled**

8. All of the following are the main features of the UDP, except ....
   (A) No connection establishment (B) No connection state at sender and receiver **(C) Large header size** small
   (D) No congestion control

9. In..............., it uses only ACKs. Instead of NAK, the receiver sends ACK for the last pkt that was received OK.
   (A) rdt2.0 **(B) rdt2.2** (C) rdt3.0 (D) rdt2.1

10. TCP congestion and flow control set window size. This is called ....
    (A) flow controlled (B) connection-oriented **(C) pipelining** (D) full duplex data

No.	Answer	No.	Answer	No.	Answer	No.	Answer	No.	Answer
1		2		3		4		5	
6		7		8		9		10	

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only ACKs → rdt 2.2

Seq. number <0,1 → rdt 2.1

Correct & corrupted → rdt 2.0

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Quiz 04 (5 Marks) - Model (4)

Choose the right answers and label them in your answer sheet (Q1-Q10 x 0.5 Mark)

1. Websites and client browsers use .... to maintain some state between transactions.
   (A) **cookies** (B) sessions (C) databases (D) Web caches
2. The three main phases of E-mail transfer are the following, except SMTP ....
   (A) handshaking (B) transfer of messages (C) **acknowledgment** (D) closure
3. .... permit sites to learn a lot about the user on their site.
   (A) Sessions (B) Databases (C) **Cookies** (D) Web caches
4. All of the following are DNS services, except ....
   (A) **Centralize hostname-to-IP-address translation database** (B) host aliasing (C) mail server aliasing (D) load distribution
5. Cookies can be used for all of the following, except ....
   (A) authorization (B) shopping carts (C) **content integrity** (D) recommendations
6. Consider sending over HTTP/2 a Web page that consists of one video clip, and five images. Suppose the video clip is split into 2000 frames, and each image consists of three frames. If all the video frames are sent first without interleaving, how many "frame times" are needed to send all five images?
   (A) 2000 (B) 2030 (C) 4030 (D) **2015**
7. In..............., demultiplexing using only the destination port number.
   (A) **UDP** (B) TCP (C) IP (D) Socket
8. The UDP segment header consists of .... bytes.
   (A) 24 (B) 16 (C) 32 (D) **20**
9. In ...., the underlying channel may flip bits in the packet. The receiver explicitly tells the sender that pkt received OK (ACKS) or pkt had errors (NAKs).
   (A) rdt2.1 (B) rdt2.2 (C) **rdt2.0** (D) rdt3.0
10. In TCP, the sender will not overwhelm the receiver. It is called ....
    (A) connection-oriented (B) pipelining (C) full duplex data (D) **flow controlled**

No.	Answer	No.	Answer	No.	Answer	No.	Answer	No.	Answer
1		2		3		4		5	
6		7		8		9		10	

The end of the examination
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Network Mid 2025 (General)

1- …….. is the time for a small packet to travel from client to server and back.
a) RTT b) HOL c) RFC d) GDPR

2- In …….., different channels transmitted in different frequency bands.
a) HFC b) FDM c) CMTS d) DSL

3- …….. define the format, order of messages sent and received among network entities, and actions taken on message transmission.
a) Routers b) Protocols c) Switches d) Servers

4- In …… transmission, order of requested objects is based on client-specified object priority. It pushes unrequested objects to the client, divides them into frames, and then schedules them.
a) HTTP 1.1 b) HTTP 1.2 c) HTTP 2.1 d) HTTP 2

5- In ……….., end-to-end resources are allocated to and reserved for a call between source and destination.
a) Queueing b) TDM c) Packet-Switching d) Circuit Switching

6- A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, another packet is halfway done being transmitted on this outbound link, and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1500 bytes and the link rate is 2.5 Mbps. What is the queueing delay for the packet?
a) 21.6 ms b) 19.2 ms c) 16.8 ms d) 2.4 ms

7- …….. attacks intercept DNS queries, returning bogus replies. bogus → Spoofing
a) ICANN b) DoS c) Spoofing d) DDoS

8- In …….., fiber attaches homes to the ISP router, and homes share access network to the cable headend.
a) DSL b) Packet Transmission Delay c) TDM d) Cable-Based Access

9- …….. provides retrieval, deletion, and folders of stored messages on the email server.
a) IMAP b) ICANN c) POP d) SMTP

10- A small object may have to wait for transmission behind a large object. This is called …….. blocking.
a) ICANN b) GDPR c) HOL d) FCFS

11- …….. is the rate (bits/time unit) at which bits are being sent from sender to receiver.
a) Throughput b) Queueing Delay c) Transmission Delay d) Nodal Processing

12- Consider sending over HTTP/2 a web page that consists of one video clip and five images. Suppose the video clip is transported as 2000 frames, and each image has three frames. If all the video frames are sent first without interleaving, how many “Frame Times” are needed until all five images are sent?

a) 2000 b) 2015 c) 2030 d) 4030

13- In …….. existing telephone lines are used to connect to a central office with a 24-52 Mbps dedicated downstream transmission rate and a 3.5-16 Mbps dedicated upstream transmission rate.

a) DSL b) TDM c) DSLAM d) CMTS

14- …….. uploads a new file (object) to the server, completely replacing a file at a specified URL with content in the entity body of the HTTP request message.

a) POST b) HEAD c) GET d) PUT

15- HTTP response status code equals to …….. when the required object is moved to a new location specified later in the message.

a) 400 b) 305 c) 301 d) 505

OK 200
moved 301
bad 400

not found 404

16- The following are fields in RR format in DNS records, except ……..

a) TTL b) URL c) Type d) Value

17- Users are generating data at a rate of 100 Kbps when busy but are busy generating data only with probability p = 0.1. Suppose that we have a 1Gbps link. What is N, the maximum number of users that can be supported simultaneously under circuit switching?

a) 10000 b) 100000 c) 1000 d) 5000

18- In …….., attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming the resource with bogus traffic.

a) Wireshark b) DoS c) Spoofing d) Sniffing

19- …….. is the premier standards development organization for the internet.

a) FDM b) HFC c) RFC d) IETF

20- SMTP uses TCP to reliably transfer email messages from client to server in port ……..

a) 80 b) 29 c) 25 d) 27

SMTP

2 25 25

Network Mid 2025 (Credit)

seg- TCP/UDP

A P S T N D P

1- What is the role of the transport layer in the OSI model?

α a) Logical addressing and routing of data packets

α b) Formatting, encryption, and compression of data

c) Reliable delivery and end-to-end communication of data

α d) Establishing, managing, and terminating sessions or connections

2- Users are generating data at a rate of 100 kbps when busy, but are busy generating data only with probability p = 0.1. Suppose that we have 1 Gbps link. What is N, the maximum number of users that can be supported simultaneously under circuit switching?

a) 10000 b) 1000 c) 5000 d) 100000

3- Which layer of the OSI model does a router operate at?

a) Data Link layer b) Transport layer c) Network layer d) Application layer

4- …… provides retrieval, deletion, and folders of stored messages on the Email server.

a) POP b) ICANN c) IMAP d) SMTP

5- What is one of the primary functions of a router?

α a) To manage network traffic

b) To examine destination IP addresses and make forwarding decisions

c) To amplify network signals

d) To connect devices to a network

6- What does the throughput rate represent in a network?

a) The highest and lowest frequencies in the network

b) The rate of incoming data at a specific point in the network

c) The capacity of the network's servers and workstations

d) The speed at which information can be sent between devices

S P F

7- What are the services provided by the data link layer? A P S T N D P

MAC error

α a) Logical addressing, routing

α b) Segmentation, reassembly, error checking

c) Framing, error detection and correction, flow control

d) Session establishment, maintenance, synchronization

8- What does IMAP stand for?

a) Internet Mail Access Protocol

b) Internet Mail Authentication Protocol

c) Internet Message Access Protocol

d) Internet Message Authentication Protocol

$L = 1500 \times 8 = 12000$, $R = 2.5 \times 10^6$

9- The following are fields in RR format in DNS records, except:
a) URL b) Type c) Value d) TTL

$t.d = L/R = \frac{12000}{2.5 \times 10^6} = 4.8 \times 10^{-3} = 4.8ms$

10- What is the advantage of a tree network topology?
a) Point-to-point wiring for individual segments device
b) Direct connectivity between devices without relying on a central device
c) Easy to install and wire
d) Scalability for future network expansion

11- A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2.5 Mbps. What is the queuing delay for the packet?
a) 19.2 ms b) 21.6 ms c) 16.8 ms d) 2.4 ms

L 4 0.5 R 4.5

$q.d = 4.8 \times 4 = 21.6ms$

12- What is the standard that defines wireless local area networks (WiFi)?
a) IEEE 802.11 b) IEEE 802.3 c) IEEE 802.16 d) IEEE 802.15

13- What is the purpose of an IP address?
a) To encrypt data during transmission b) To manage network connections
c) To identify a computer or network device on the Internet d) To establish secure tunnels

14- ...... attacks intercept DNS queries, returning bogus replies.
a) DDoS b) ICANN c) Spoofing d) DoS

Spoofing

15- What type of model does HTTP follow?
a) Peer-to-peer model b) Client-server model c) Broadcast model d) Hybrid model

16- Which protocols are commonly used in the network layer?
a) Ethernet, Point-to-Point Protocol (PPP), Wi-Fi
b) Transmission Control Protocol (TCP), User Datagram Protocol (UDP)
c) HTTP, FTP, SMTP
d) Internet Protocol (IP)

17- Consider sending over HTTP/2 a Web page that consists of one video clip, and five images. Suppose that the video clip is transported as 2000 frames, and each image has three frames. If all the video frames are sent first without interleaving, how many "frame times" are needed until all five images are sent?
a) 2000 b) 2030 c) 4030 d) 2015

$2000 + 15 = 2015$

15 frame

18- UDP is suitable for applications that prioritize
a) Low latency and real-time data transmission b) Reliability
c) Flow control d) Ordered data transmission

19- Which layer handles the formatting, encryption, and compression of data in the OSI model?
a) Physical layer b) Presentation layer c) Data link layer d) Session layer

20- Which network device is responsible for assigning IP addresses to devices on a network?
a) DNS server b) DHCP server c) File server d) Web server

Oral Network 2025

1- What is the definition of Routing queuing delay?

$\frac{L}{R} \times$ no. of packets in queue

2- What is the RFC?

3- What is IETF?

International organize to put internet standards.

4- What is the difference between HTTP/1.1, HTTP/2 , HTTP/3?

FCFS

object priority

more secured

5- What is the difference between Forwarding and Routing?

using Routing protocols

6- What is the difference between DoS and Sniffing and Spoofing?

7- What is the main feature of Vanilla TCP?

8- What is TLS?

Network Final 2025 (تخلفات)

S P F
A P S T N D P
MAC error

Part I: Choose the right answer (40 x 1.25 Marks)

1. Which network device operates at the data link layer of the OSI model?
(A) Router (B) Switch (C) Hub (D) Modem

2. What are the services provided by the data link layer?
(A) Framing, error detection and correction, flow control
(B) Logical addressing, routing
(C) Segmentation, reassembly, error checking
(D) Session establishment, maintenance, synchronization

3. What is the advantage of a tree network topology?
(A) Point-to-point wiring for individual segments
(B) Direct connectivity between devices without relying on a central device
(C) Easy to install and wire
(D)Scalability for future network expansion

4. What is the default port for FTP?
(A) 21 (B) 25 (C) 53 (D) 80

5. What is the primary advantage of wireless networks for mobile users?
(A) Faster data transmission (B) Accessibility and convenience
(C) Greater security (D) Lower cost

6. Which type of network involves the establishment of a connection before transinitting data?
(A) Packet Switching (B) Half Duplex (C) Full Duplex (D) Circuit Switching

7. Which protocol supports packcts containing error messages, control, and information?
(A) ARP (B)UDP (C) ICMP (D) RARP

8. Which type of trausmission media is resistant to interference and supports long cable leugths?
(A) Shielded twisted pair (STP) Cable (B) Unshielded twisted pair (UTP) Cable
(C) Coaxial Cable (D) Fiber Optic Cable

9. What is the measurement unit used for bandwidth?
(A) Mbps (B) GB (C) MHZ (D) KB

10. Which layer of the OSI model is responsible for error delection and correction?
(A) Physical layer (B) Network layer (C) Data link layer (D) Transport layer

A P S T N D P
MAC error

11. What is the standard number for Token Ring technology defined by the IEEE?
(A) 802.3 (B) 802.11 (C) 802.15 (D) 802.5

12. Which protocol is commonly used by routers to exchange routing information?
(A) TCP (B) UDP (C) RIP (D) IP
13. What are the four phases of DHCP communication known as?
(A) ABCD (B) DORA (C) DHCP (D) IPAD

14. What is one of the primary functions of a router?
(A) To manage network traffic
(B) To examine destination IP addresses and make forwarding decisions
(C) To amplify network signals
(D) To connect devices to a network

15. Which type of cable is used to connect a router to a switch or hub in Ethernet networks?
(A) Rolled cable (B) Crossover cable (C) Straight-through cable (D) all the Above

16. What is the purpose of subnetting in a network?
(A) To extend the geographical coveruge of a network
(B) To establish secure VPN connections
(C) To allocate IP addresses to devices
(D) To divide a large network into smaller logical networks

17. Which type of cable is used to connect a hub/switch to another hub/switch in Ethernet networks?
(A) Rolled cable (B) Straight-through cable (C) Unshiekled twisted pair (UTP) Cable
(D) Crossover calle

18. Which protocol is used for retrieving emails from a mail server?
(A) SMTP (B) POP3 (C) IMAP (D) FTP

19. Which type of network card is more popular than Token Ring?
(A) Ethernet (B) Coaxial (C) BNC (D) RJ-45

20. Which layer handles the formatting, encryption, and compression of data in the OSI model?
(A) Physical layer (B) Presentation layer (C) Data link layer (D) Session layer

21. What is the purpose of DNS in a computer network?
(A) To convert domain names to IP addresses (B) To encrypt data transmission
(C) To manage network security (D) To establish wireless connections

22. Which wireless standard provides the highest data transfer rate?
(A) 802.11a (B) 802.11b (C) 802.11g (D) 802.11ac

23. Which IP address class is used for multicast?
(A) Class A (B) Class B (C) Class C (D) Class D

24. What does the throughput rate represent in a network?
(A) The highest and lowest frequencies in the network
(B) The rate of incoming data at a specific point in the network
(C) The capacity of the network's servers and workstations
(D) The speed at which information can be sent between devices

25. What is the purpose of CIDR notation?
(A) To represent IP addresses in binary format
(B) To indicate the number of available host addresses in a network
(C) To separate the network prefix from the host portion of an IP address
(D) To indicate the class of an IP address

26. What is the purpose of the OUI in MAC addresses?
(A) To identify the IP address of the device
(B) To determine the network segment of the device
(C) To identify the manufacturer of the network equipment
(D) To encrypt the data transmitted over the network

27. Which network topology provides the highest level of fault tolerance?
(A) Star topology (B) Bus topology (C) Ring topology (D) Mesh topology

28. In which mode of operation does data flow in both directions alternately?
(A) Half Duplex (B) Full Duplex (C) Simplex (D) Broadcast

29. What is the purpose of NAT (Network Adress Translation)?
(A) To translate domain names to IP addresses
(B) To map private IP addresses to public IP addresses
(C) To encrypt network traffic
(D) To assign IP addresses to devices

30. Which mode of operation involves the output of data from the sender being received by all other receivers?
(A) Simplex (B) Broadcast (C) Half Duplex (D) Full Duplex

31. What device is used to amplify the strength of a network signal?
(A) Repeater (B) Bridge (C) Router (D) Switch

32. Which protocol is used for transferring web pages from a web server to a client browser?
(A) FTP (B) SMTP (C) HTTP (D) POP3

33. Which IP addressing method allows for variable-length subnet masks?
(A) Classful addressing (B) Classless adaressing
(C) Class-based addressing (D) Supernetting

34. Which network devices are typically involved in packet forwarding?
(A) Servers and workstations (B) Routers and switches
(C) Modems and gateways (D) Firewalls and bridges

35. Which protocols are commonly used in the network layer?
(A) Ethernet, Point-to-Point Protocol (PPP), Wi-Fi
(B) Internet Protocol (IP)
(C) Ethernet, Point-to-Point Protocol (PPP), Wi-Fi
(D) HTTP, FTP, SMTP

MAC

36. Which protocol is used to determine the physical address of a host on the network?
(A) RARP (B) ARP (C) ICMP α (D) UDP

37. Which layers of the OSI model are responsible for the communications between network devices?
(A) Layers 1-3 (B) Layers 2-4 (C) Layers 3-5 (D) Layer 1

38. Which type of transmission media is suitable for environments with electrical noise and interference?
(A) Coaxial Cable (B) Shielded twisted pair (STP) Cable
(C) Unshielded twisted pair (UTP) Cable (D) Fiber Optic Cable

39. Which IP address range is reserved for private networks?
(A) 10.0.0.0 - 10.255.255.255 (B) 172.16.0.0 - 172.31.255.255
(C) 192.168.0.0 - 192.168.255.255 (D) All of the above

40. Which network device is responsible for assigning IP addresses to devices on a network?
(A) DNS server α (B) File server α (C) DHCP server (D) Web server α

Part II: Answer the following question (10 Marks)

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28.
(a) What is the first address in the block?
(b) What is the last address in the block?

/28

$32-28=4$ $2^4=16$

$39-(39\ \%\ 16)$
$=39-7=32$

205.16.37.32 → 1st address
205.16.37.47 → last address

Network Final 2025 (Credit)

1. ...... topology is used to interconnect the routers in the network.
   (A) Mesh (B) Bus (C) Ring (D) Star

2. ...... is commonly used in traditional telephone networks.
   (A) Packet switching (B) Circuit switching (C) Routing (D) Forwarding

3. ...... protocol is responsible for datagram format, addressing, and packet handling conventions.
   (A) OSPF (B) IP (C) BGP (D) ICMP

4. Check bit errors and determination output link, it may typically take less than microseconds.
   (A) Transmission delay (B) Queueing delay (C) Codal processing
   (D) Propagation delay

5. In the router, longest prefix matching is often performed using ...... it retrieves addresses in one clock cycle, regardless of table size.
   (A) CAM (B) RED (C) ECN (D) TCAM

6. ...... method includes user data in the URL, field of the HTTP request message following a '?'.
   (A) GET (B) HEAD (C) PUT (D) POST

7. The non-persistent HTTP response time equals to ......
   (A) As little as one RTT for all the referenced objects
   (B) As little as two RTT for all the referenced objects
   (C) 2 RTT (D) 2 RTT + file transmission time

8. Users are generating data at a rate of 100kbps when busy, but are busy generating data only with probability p = 0.1, suppose that we have 1 Gbps link. What is N, the maximum number of users that can be supported simultaneously under circuit switching?
   (A) 10000 (B) 1000 (C) 5000 (D) 100000

9. In ......, the underlying channel can have errors and lose packets (data, ACKs).
   (A) Rdt2.2 (B) Rdt2.0 (C) Rdt3.0 (D) Rdt2.1

10. ...... handles data from multiple sockets and adds a transport header.
    (A) Multiplexing (B) Demultiplexing (C) Routing (D) Forwarding

11. In ......, a promiscuous network interface reads/records all packets passing by.
(A) IP spoofing (B) DoS (C) Packet ~~stuffing~~ (D) Malware
sniffing

12. The three major components of E-mail are the following, except......
(A) User agents (B) IMAP (C) Mail servers (D) SMTP

13. ...... is a standards organization for the internet and is responsible for the technical standards that make up the internet protocol suite (TCP/IP).
(A) RFC (B) IETF (C) ISP (D) CMTS

14. ...... is an example service for individual datagrams.
(A) Guaranteed delivery with less than 40 msec delay
(B) In-order datagram delivery (C) Guaranteed minimum bandwidth to flow
(D) Restrictions on changes in inter-packet spacing

15. ...... is the main component of the network core that concerns the network interconnection.
(A) Server (B) Host (C) Hub (D) Router

16. For the DiffServ service model, the quality of service guarantees the following value for bandwidth, loss, order, and timing respectively.
(A) Guaranteed min, no, yes, no (B) Constant rate, yes, yes, yes
(C) Possible, possibly, possibly, no (D) Yes, yes, yes, yes

17. In the network, all of the following are lines of defense, except ......
(A) Encryption (B) Gateway (C) Integrity checks (D) Firewalls

18. In ...... it handles garbled ACK/NAKs by adding a sequence number to the packet.
(A) Rdt2.0 (B) Rdt2.1 (C) Rdt2.2 (D) Rdt3.0

19. In the network layer, ... plane determines how the datagram is routed among routers along the end-end path from source host to destination host.
(A) Data (B) Control (C) Service (D) Information

20. The main difference that distinguishes HTTP/3 over the other earlier versions is ......
(A) Transmission order of requested objects based on client-specified object priority
(B) The server responds in order (FCFS) to GET requests
(C) The connection is persistent
(D) It adds security, per object error-and congestion control (more pipelining) over UDP

21. In ...... control plane, individual routing algorithm components in every router interact in the control plane.
(A) SDN (B) DSL (C) QoS (D) Per-router

22. ...... is the time for a small packet to travel from client to server and back.
(A) Internet delay (B) RTT (C) Access link delay (D) LAN delay

23. ...... manages root DNS domain.
(A) ICANN (B) DNSSEC (C) Comcast (D) Akamai

24. UDP can be used for all of the following except ......
(A) DNS (B) HTTP/3 (C) SNMP (D) IMAP

25. All communication activity on the internet is governed by ......
(A) Services (B) ISP (C) Gateway (D) Protocols

26. ...... process initiates communication.
(A) P2P (B) Server (C) Inter (D) Client

27. ...... moves arriving packets from the router's input link to the appropriate router output link.
(A) Routing (B) Multiplexing (C) Demultiplexing (D) Forwarding

28. Cable is two concentric copper conductors that can work in bidirectional with multiple frequency channels on cable.
(A) Coaxial (B) Fiber optic (C) Twisted pair category 5 (D) Twisted pair category 6

29. ...... is a connection between the host/router and the physical link.
(A) Interface (B) Subnet (C) NAT (D) WFQ

$5 \times 3 = 15$

30. Consider sending over HTTP/2 a web page that consists of one video clip, and five images, suppose that the video clip is transported as 2000 frames, and each image has three frames, if all the video frames are sent first without interleaving how many "frame times" are needed until all five images are sent?
(A) 2000 (B) 2030 (C) 4030 (D) 2015

video = 2000
15
2015

31. ...... is a time waiting at the output link for transmission, it depends on the congestion level of the router.
(A) Transmission delay (B) Nodal processing (C) Propagation delay
(D) Queueing delay

32. In the router switching via ...... it fragments the datagram into fixed-length cells on entry, it switches cells through the fabric and reassembles the datagram at the exit.
(A) Memory (B) Bus (C) Interconnection network (D) Path

reliable

33. In ...... demultiplexing uses 4 tuple: source and destination IP addresses and port numbers.
(A) UDP (B) IP (C) TCP (D) Socket

Part II: answer the following question (10 Marks)

A. What is congestion? What are the two common approaches towards congestion control (write the main features)?

B. What are the main advantages and disadvantages of static routing?